amazon

BRAZIL E-COMMERCE ANALYTICS

Comprehensive SQL Data & Business Analysis Project

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Database Source: Amazon Brazil E-Commerce Marketplace

Date: July 2026

Executive Summary & Introduction

This comprehensive technical report delivers data-driven operational insights into the Amazon Brazil marketplace. Utilizing structural database management systems and complex SQL architectures, the dataset analyzes commercial performance, logistics, consumer sentiment, transactional behaviors, and seller distributions.

The core objective of this project is to turn raw transactional databases into actionable corporate strategies. By answering 14 core operational questions, this analysis provides high-level metrics for standardizing operations, targeting marketing efforts, optimizing geographic distributions, and driving regional revenue growth.

Q1: Seller Distribution and Order Fulfillment by State

Question Formulation

Find the total number of orders fulfilled by each seller state across the Amazon Brazil marketplace network.

SQL Query Architecture

```
SELECT
    s.seller_state,
    COUNT(DISTINCT oi.order_id) AS total_orders
FROM sellers s
JOIN order_items oi
    ON s.seller_id = oi.seller_id
GROUP BY s.seller_state
ORDER BY total_orders DESC;
```

Query Output Data Matrix

Seller State	Total Orders Fulfilled
SP (São Paulo)	70,188
MG (Minas Gerais)	7,930
PR (Paraná)	7,673
RJ (Rio de Janeiro)	4,353
SC (Santa Catarina)	3,667
RS (Rio Grande do Sul)	1,989
DF (Distrito Federal)	824
BA (Bahia)	569
GO (Goiás)	463
PE (Pernambuco)	406
MA (Maranhão)	392
ES (Espírito Santo)	318
MT (Mato Grosso)	137
CE (Ceará)	91
RN (Rio Grande do Norte)	51
MS (Mato Grosso do Sul)	49
PB (Paraíba)	36
RO (Rondônia)	14
PI (Piauí)	12
SE (Sergipe)	9
PA (Pará)	4
AM (Amazonas)	3
AC (Acre)	1

BUSINESS ANALYST STRATEGIC INSIGHTS

Geographic Concentration: The structural distribution reveals massive operational centralization in São Paulo (SP), accounting for over 70,000 orders. This constitutes the vast majority of all marketplace transactions, pointing to a severe economic centralization within the Southeast region of Brazil.

Strategic Action Items:

- **Supply Chain Safety:** Relying heavily on a single hub like SP introduces supply chain risks. If a localized disruption occurs, the entire network will experience delays.
- **Regional Opportunities:** States like MG and PR show strong baseline performance. Amazon can invest in regional seller onboarding programs to reduce shipment distances and lower inter-state freight costs.

Q2: Cumulative Revenue Trajectory by Product Category

Question Formulation

For each product category, calculate the cumulative revenue generated as orders come in chronologically over time.

SQL Query Architecture

```
SELECT
    p.product_category_name,
    DATE(o.order_purchase_timestamp) AS order_date,
    SUM(oi.price) AS daily_revenue,
    SUM(SUM(oi.price)) OVER (
        PARTITION BY p.product_category_name
        ORDER BY DATE(o.order_purchase_timestamp)
    ) AS cumulative_revenue
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
JOIN orders o ON oi.order_id = o.order_id
GROUP BY p.product_category_name, DATE(o.order_purchase_timestamp);
```

Query Output Data Matrix (Sample Time Series Data)

Product Category Name	Order Date	Daily Revenue (BRL)	Cumulative Revenue (BRL)
beleza_saude	2016-10-04	49.99	49.99
beleza_saude	2016-10-08	15.54	65.53
beleza_saude	2017-01-17	169.36	234.89
beleza_saude	2017-01-20	319.99	554.88
beleza_saude	2017-01-25	378.90	933.78
beleza_saude	2017-01-26	33.90	967.68
beleza_saude	2017-01-27	602.72	1,570.40
beleza_saude	2017-01-28	84.86	1,655.26
beleza_saude	2017-01-30	90.00	1,745.26
beleza_saude	2017-01-31	23.26	1,768.52

BUSINESS ANALYST STRATEGIC INSIGHTS

Revenue Growth Trends: Tracking chronological revenue acceleration allows the finance team to monitor product life cycles and sales velocity. The steep upward slope shows that health and beauty products have consistent, recurring buyer patterns.

Strategic Action Items:

- **Inventory Management:** Use this cumulative run-rate to forecast working capital and inventory requirements across multiple fulfillment centers.
- **Dynamic Ad Placement:** Increase advertising spending during windows that show high growth acceleration to maximize your return on ad spend (ROAS).

Q3: Consumer Payment Preference & Average Transaction Size

Question Formulation

Identify the payment methods customers use most frequently, alongside the average order value associated with each unique type.

SQL Query Architecture

```
SELECT
    payment_type,
    COUNT(*) AS total_transactions,
    ROUND(AVG(payment_value), 2) AS average_order_value
FROM order_payments
GROUP BY payment_type
ORDER BY total_transactions DESC;
```

Query Output Data Matrix

Payment Type	Total Transactions	Average Order Value (BRL)
credit_card	76,795	163.32
boleto	19,784	145.03
voucher	5,775	65.70
debit_card	1,529	142.57
not_defined	3	0.00

BUSINESS ANALYST STRATEGIC INSIGHTS

Dominance of Credit Financing: Credit cards are the primary choice for consumers, leading both in total volume (76k+ transactions) and average order value (163.32 BRL). Interestingly, Boleto Bancário remains a highly relevant secondary option in Brazil, capturing nearly 20,000 transactions due to cash-based shopping preferences.

Strategic Action Items:

- **Flexible Installments:** Offering attractive credit installment plans (parcelamento) can help increase your average basket value.
- **Boleto Optimizations:** Since boletos take longer to process, implementing fast digital boleto or Pix updates can speed up confirmation times and shorten your shipping cycles.

Q4: Identification of Top Tier VIP High-Net-Worth Consumer

Question Formulation

Isolate and extract the customer profile that has spent the highest amount of money cumulative across all platform orders.

SQL Query Architecture

```
SELECT
    o.customer_id,
    SUM(op.payment_value) AS total_spent
FROM orders o
JOIN order_payments op ON o.order_id = op.order_id
GROUP BY o.customer_id
ORDER BY total_spent DESC
LIMIT 1;
```

Query Output Data Matrix

Customer Unique Identification Token	Total Cumulative Capital Spent (BRL)
1617b1357756262bfa56ab541c47bc16	13,664.08

BUSINESS ANALYST STRATEGIC INSIGHTS

High-Value Customer Profiles: A single customer account spending over 13.6k BRL highlights the impact of high-value shoppers. In e-commerce, a small group of VIP users often drives a large portion of overall profitability.

Strategic Action Items:

- **VIP Loyalty Support:** Design a premium retention program that offers exclusive rewards, early access to new products, and dedicated account support for top-tier spenders.
- **Personalized Marketing:** Build recommendation engines tailored to high-spending behaviors to encourage repeat purchases in high-margin categories.

Q5: Qualitative Assessment of Product Categories via Customer Reviews

Question Formulation

Compute the arithmetic average user review score for each product category on the platform to identify quality thresholds.

SQL Query Architecture

```
SELECT
    p.product_category_name,
    ROUND(AVG(r.review_score), 2) AS average_review_score
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
JOIN order_reviews r ON oi.order_id = r.order_id
GROUP BY p.product_category_name
ORDER BY average_review_score DESC;
```

Query Output Data Matrix (Sample Selection)

Product Category Name	Average Review Score (Scale 1-5)
cds_dvds_musicals	4.50
fashion_roupas_infanto_juvenil	4.50
livros_importados	4.44
livros_tecnicos	4.42
construcao_ferramentas_ferramentas	4.40
alimentos_bebidas	4.37
malas_acessorios	4.32
portateis_casa_forno_cafe	4.32
fashion_esporte	4.30
fashion_calcados	4.26
brinquedos	4.21
papelaria	4.18
perfumaria	4.17

BUSINESS ANALYST STRATEGIC INSIGHTS

Quality Metrics: Niche categories like media and specialized books score highly (4.4 - 4.5), indicating that these products consistently meet customer expectations. Mass-market categories like stationery and perfumes score slightly lower, suggesting potential issues with product matching or shipping conditions.

Strategic Action Items:

- **Vendor Auditing:** Review categories with lower satisfaction ratings to identify underperforming sellers or common product issues.
- **Enhanced Product Pages:** Introduce clearer descriptions, compatibility guides, and size charts to lower returns and improve satisfaction.

Q6: Customer Engagement and Granular Regional Purchase Frequency

Question Formulation

Find the total number of orders placed by each unique customer, broken down by their resident home state.

SQL Query Architecture

```
SELECT
    c.customer_state,
    c.customer_id,
    COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_state, c.customer_id
ORDER BY total_orders DESC;
```

Query Output Data Matrix (Sample Distribution Matrix)

- ... All customers track cleanly against their regional metrics.

Customer State	Customer Identification Key Token	Total Orders Placed
SP	c9cbdbc1e75330a706634b7110b36b51	1
MG	ff7b309f678ba3fc6bc7093c6fb5c721	1
BA	73a75b9ae754b29d67d6be7bdad827e5	1
MG	19b4079d0b4345674c8a30f53709fc46	1

BUSINESS ANALYST STRATEGIC INSIGHTS

Order Patterns: The data shows that many buyer accounts currently register a single transaction. This highlights a classic industry challenge: moving customers from their first purchase to a habit of repeat shopping.

Strategic Action Items:

- **Automated Retargeting:** Set up automated follow-up emails with tailored discounts to encourage a second purchase within 30 days of the first order.
- **Loyalty Incentives:** Launch a subscription model or standard loyalty program to increase purchase frequency and build long-term retention.

Q7: Identification of Zero-Yield Inactive Merchant Registrations

Question Formulation

Identify registered merchants on the platform who have never completed or fulfilled a single customer transaction.

SQL Query Architecture

```
SELECT
    s.seller_id,
    s.seller_state
FROM sellers s
LEFT JOIN order_items oi ON s.seller_id = oi.seller_id
WHERE oi.order_id IS NULL;
```

Query Output Data Matrix

Seller ID Token	Seller Base State Location
[Empty Dataset / All current sellers actively processing transactions]	

BUSINESS ANALYST STRATEGIC INSIGHTS

Seller Engagement: An empty result set indicates excellent health for the seller network: every registered seller has successfully completed at least one order. This shows low friction in the merchant onboarding process.

Strategic Action Items:

- **Proactive Support:** Maintain this high engagement by monitoring new sellers during their first 90 days to help them secure their first sale quickly.
- **Platform Health Metrics:** Continue tracking this metric quarterly to clean up inactive accounts and optimize platform performance.

Q8: High Margin Strategic Top 5 Revenue-Generating Categories

Question Formulation

Identify the top 5 product categories that generate the highest total revenue across the entire platform.

SQL Query Architecture

```
SELECT
    p.product_category_name,
    SUM(oi.price) AS total_revenue
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 5;
```

Query Output Data Matrix

Product Category Strategic Classification	Total Financial Revenue Generated (BRL)
beleza_saude (Health & Beauty)	1,258,681.34
relogios_presentes (Watches & Gifts)	1,205,005.68
cama_mesa_banho (Bed, Bath & Table)	1,036,988.68
esporte_lazer (Sports & Leisure)	988,048.97
informatica_acessorios (Computers Accessories)	911,954.32

BUSINESS ANALYST STRATEGIC INSIGHTS

Core Growth Areas: The top categories are led by Health & Beauty and Watches & Gifts, both generating over 1.2 million BRL. These sectors are critical revenue drivers for the platform.

Strategic Action Items:

- **Supplier Partnerships:** Strengthen relationships with key brands in these categories to secure exclusive deals and protect profit margins.
- **Targeted Marketing Campaigns:** Allocate a larger share of your marketing budget to these high-performing categories during seasonal shopping events.

Q9: Marketplace Logistics Performance & Median Delivery Window

Question Formulation

Calculate the median delivery turnaround time in days for all completed customer orders.

SQL Query Architecture

```
WITH delivery_time AS (
    SELECT DATEDIFF(order_delivered_customer_date, order_purchase_timestamp) AS
delivery_days
    FROM orders
    WHERE order_delivered_customer_date IS NOT NULL
)
SELECT AVG(delivery_days) AS median_delivery_days
FROM (
    SELECT delivery_days,
           ROW_NUMBER() OVER (ORDER BY delivery_days) AS rn,
           COUNT(*) OVER () AS total_rows
    FROM delivery_time
) t
WHERE rn IN ((total_rows + 1)/2, (total_rows + 2) / 2);
```

Query Output Data Matrix

Logistical Metric Name	Calculated Value (Days)
Median Delivery Turnaround Window	10.0000 Days

BUSINESS ANALYST STRATEGIC INSIGHTS

Logistics Benchmarking: A median delivery time of 10 days provides a stable baseline for operations. However, in modern e-commerce, long shipping times can lower customer retention rates.

Strategic Action Items:

- **Fulfillment Centers:** Set up regional fulfillment hubs closer to high-demand areas to reduce delivery times down to 2-4 days.
- **Carrier Performance:** Review delivery partners in slower regions to improve transit times and offer more reliable service.

Q10: Audit of Zero-Velocity Unordered Product Inventory

Question Formulation

Identify listed products in the inventory catalog that have never been purchased by a customer.

SQL Query Architecture

```
SELECT
    p.product_id,
    p.product_category_name
FROM products p
LEFT JOIN order_items oi ON p.product_id = oi.product_id
WHERE oi.product_id IS NULL;
```

Query Output Data Matrix

Product Unique Identifier Code	Product Category Classification
[Empty Dataset / All catalog inventory active]	

BUSINESS ANALYST STRATEGIC INSIGHTS

Inventory Health: The clean result indicates there is no stagnant inventory in the catalog. Every listed item has generated at least one sale, showing high catalog efficiency.

Strategic Action Items:

- **Dynamic Recommendations:** Continue using recommendation engines to feature slower-moving items before they become stagnant.
- **Catalog Reviews:** Keep monitoring catalog health to support listing quality and ensure smooth search discovery.

Q11: High-Performance Merchant Outlier Benchmarking Analysis

Question Formulation

Identify sellers whose individual order fulfillment count is higher than the average order volume across all sellers on the platform.

SQL Query Architecture

```
SELECT
    seller_id,
    COUNT(order_id) AS total_orders
FROM order_items
GROUP BY seller_id
HAVING COUNT(order_id) > (
    SELECT AVG(order_count)
    FROM (
        SELECT COUNT(order_id) AS order_count
        FROM order_items
        GROUP BY seller_id
    ) avg_orders
);
```

Query Output Data Matrix (Sample Selection)

Seller Unique Identifier Key	Total Processed Orders
48476dade18acb2bce05ec2a04120151	1,203
dd7ddc0f6c2f614352b383ee2036c143	229
7c40319990401114347954364617be29	306
15560211a19b479923666c44a7e9400a	2,033

BUSINESS ANALYST STRATEGIC INSIGHTS

Top Performing Sellers: A small group of high-volume sellers handles thousands of transactions, acting as key partners for the platform's stability.

Strategic Action Items:

- **Partner Programs:** Offer these high-volume sellers reduced platform fees or priority support to build strong, long-term partnerships.
- **Capacity Monitoring:** Track their inventory and fulfillment capabilities regularly to prevent bottlenecks during peak shopping seasons.

Q12: Regional Consumer Satisfaction Matrix by State Location

Question Formulation

Calculate the average user review score across all completed orders, grouped by the customer's state.

SQL Query Architecture

```
SELECT
    c.customer_state,
    ROUND(AVG(r.review_score), 2) AS average_review_score
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_reviews r ON o.order_id = r.order_id
GROUP BY c.customer_state
ORDER BY average_review_score DESC;
```

Query Output Data Matrix (Sample Selection)

Customer State Location	Average Review Score (Scale 1-5)
AP	4.25
TO	4.22
SP	4.14
RS	4.12
MG	4.12
MT	4.10
BA	4.04
RJ	3.92

BUSINESS ANALYST STRATEGIC INSIGHTS

Regional Satisfaction Trends: Amapá (AP) and Tocantins (TO) show high satisfaction scores, while Rio de Janeiro (RJ) drops to 3.92. This lower score in RJ is often linked to longer transit times or higher regional delivery challenges.

Strategic Action Items:

- **Targeted Operations:** Work closely with regional delivery partners in lower-scoring states to improve transit reliability and speed.
- **Localized Marketing:** Launch regional promotions and service adjustments in areas with high satisfaction to drive further growth.

Q13: Analysis of Non-Responding Customer Feedback Gaps

Question Formulation

Identify customers who completed a purchase but did not submit a product review.

SQL Query Architecture

```
SELECT DISTINCT
    c.customer_id
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
LEFT JOIN order_reviews r ON o.order_id = r.order_id
WHERE r.review_id IS NULL;
```

Query Output Data Matrix (Sample Unique Tokens)

Customer Unique Token Identification Field
9b8ce0036653563dead4613ef426b4b1
befe184e4eb181470006381368809301
482270f76d3e23bc81fc1351342f5ffa
043615210630fa1eaba81abfeab06030

BUSINESS ANALYST STRATEGIC INSIGHTS

Feedback Gaps: Customers who don't leave reviews represent a missed opportunity for gathering product data. Increasing review participation helps build trust for future buyers.

Strategic Action Items:

- **Post-Purchase Follow-Ups:** Send simple, well-timed review reminders with direct links to make sharing feedback quick and easy.
- **Review Incentives:** Offer small rewards, like loyalty points or discounts on future orders, to encourage more customers to leave reviews.

Q14: Peak Seasonality and Maximum Order Volume Month

Question Formulation

Identify the calendar month that records the highest number of completed customer orders.

SQL Query Architecture

```
SELECT
    MONTHNAME(order_purchase_timestamp) AS month_name,
    COUNT(order_id) AS total_orders
FROM orders
GROUP BY MONTH(order_purchase_timestamp), MONTHNAME(order_purchase_timestamp)
ORDER BY total_orders DESC
LIMIT 1;
```

Query Output Data Matrix

Peak Calendar Month	Total Orders Aggregated
August	10,843 Orders

BUSINESS ANALYST STRATEGIC INSIGHTS

Seasonal Trends: August emerges as the top-performing month with over 10,800 orders, signaling a key sales window driven by late-winter promotions and Father's Day shopping in Brazil.

Strategic Action Items:

- **Staffing & Logistics:** Increase warehouse staffing and courier support ahead of August to handle the seasonal surge smoothly.
- **Marketing & Inventory:** Launch targeted promotional campaigns early and stock up on high-demand items to make the most of this peak shopping period.